

UNIVERSITY OF ASIA PACIFIC

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Course Name: Operating System Lab

Course Code: CSE 402

LAB REPORT

Lab Report Number : 08

Lab Report Name: Implementation of FIFO in Page Replacement Algorithm

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1. Problem Statement:

An application is to be developed to implement the **First-In, First-Out (FIFO) Page Replacement Algorithm** using Python. The program takes a reference string of pages 1, 2, 3, 4, 2, 3, 1 and uses three available memory frames to simulate page replacement. Whenever a requested page is not present in memory, a page fault occurs. If all frames are occupied, the page that entered memory first is replaced with the newly requested page. The program displays the memory contents after each page reference and calculates the total number of page faults and page hits.

2. Objectives:

The main objectives of this laboratory experiment are:

1. To understand the concept and working principle of the First-In, First-Out (FIFO) Page Replacement Algorithm.
2. To implement the FIFO Page Replacement Algorithm using Python.
3. To simulate page loading and replacement using a given reference string and a fixed number of memory frames.
4. To calculate the total number of page faults and page hits during memory access.
5. To analyze how FIFO replaces the oldest page when all available memory frames are occupied.
6. To understand the role of page replacement algorithms in operating system memory management.

3. Souce Code Screenshot:

```python
pages = [1, 2, 3, 4, 2, 3, 1]
frames = 3

memory = []
page_faults = 0

for page in pages:
    if page not in memory:
        page_faults += 1

        if len(memory) < frames:
            memory.append(page)
        else:
            memory.pop(0)
            memory.append(page)

    print(page, "->", memory)

print("Total Page Faults:", page_faults)
print("Total Page Hits:", len(pages) - page_faults)
```

Figure 1: Source Code of the FIFO in Page Replacement Algorithm

4. Output Screenshot:

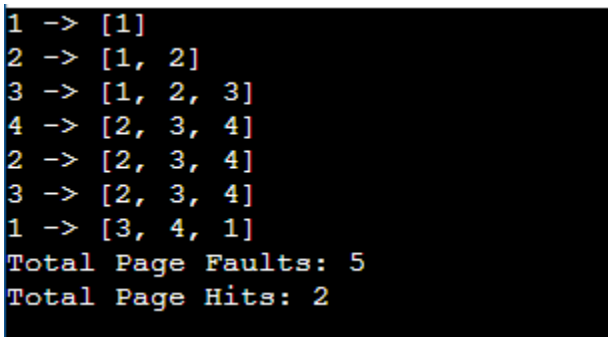


Figure 2: Execution Output of the FIFO in Page Replacement Algorithm

5. Discussion:

The FIFO (First-In, First-Out) Page Replacement Algorithm was successfully implemented using Python. The program used the reference string 1, 2, 3, 4, 2, 3, 1 with three available memory frames to demonstrate the page replacement process. Initially, pages 1, 2, and 3 were loaded into memory, resulting in three page faults. When page 4 was requested, the oldest page, page 1, was replaced according to the FIFO principle. Later, pages 2 and 3 were already present in memory, resulting in two page hits. Finally, when page 1 was requested again, page 2 was replaced because it was the oldest page in memory. The program produced a total of **5 page faults and 2 page hits**. This experiment demonstrates that FIFO is simple to implement and follows a queue-based replacement policy. However, it does not consider how frequently or recently a page is used, which may lead to unnecessary page replacements in some situations.

6. Conclusion:

The First-In, First-Out (FIFO) Page Replacement Algorithm was successfully implemented using Python. The program processed the given reference string using three memory frames and correctly displayed the memory status after each page reference. The final result showed a total of **5 page faults and 2 page hits**. This experiment helped us understand how FIFO manages memory by replacing the page that entered memory first when all available frames are occupied. Although FIFO is simple and easy to implement, it may cause unnecessary page faults because it does not consider the frequency or recent usage of pages. Therefore, the experiment provided practical knowledge of page replacement and its importance in operating system memory management.